Technology-Enabled Medication Verification System for Drug Dispensing system

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CDE4301 Project Code: EIM-404

Prototype Name

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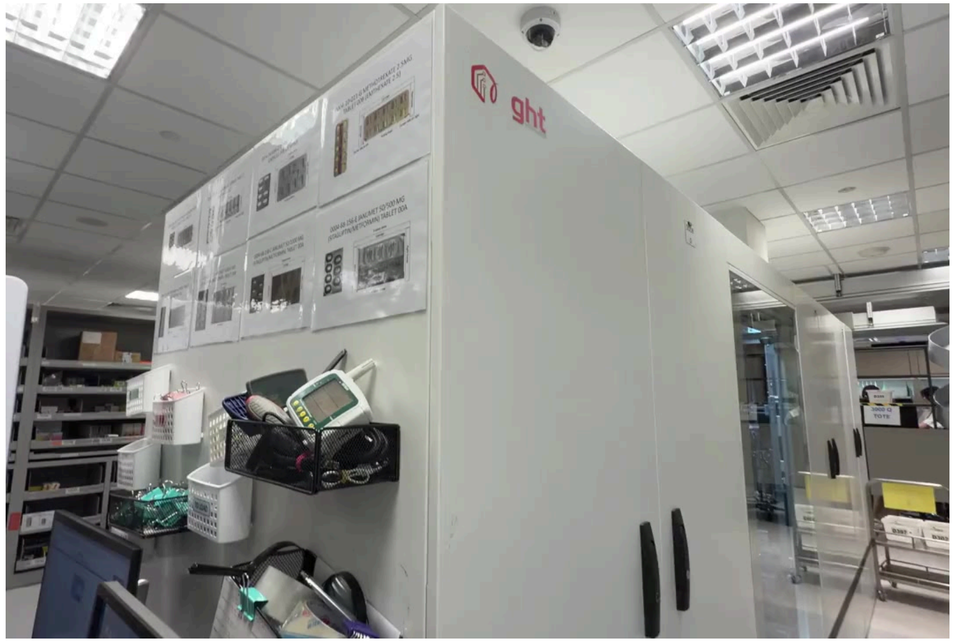
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Context

What is DDS-2 System ?

DDS = Drug Dispensing System







I feel burnout after counting 10 cartridges!

Linda has been trained and has worked for 3 years since the DDS-2 system was deployed.

She has to MANUALLY count, read, and verify the blister information to be keyed in for over 50 cartridges every day.

Till now, she has accumulated 60 errors.

This has happened to her other colleagues too...

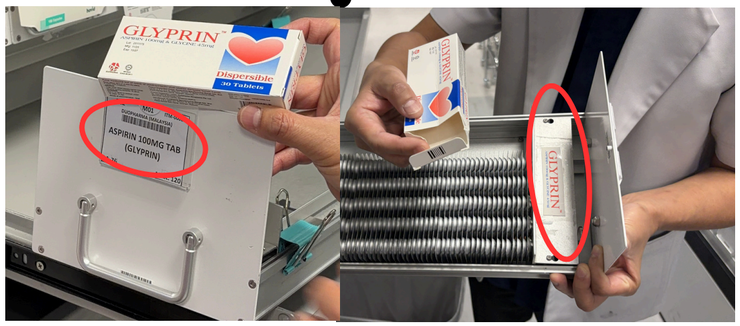
**AI generated image for illustration only

User Journey Map



Retrieve required quantities

Check labels within cartridge

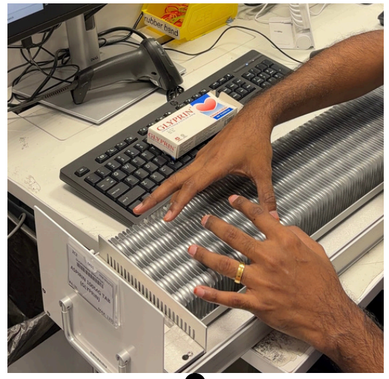




Align strips in right orientation and count concurrently

Read and key in blister information





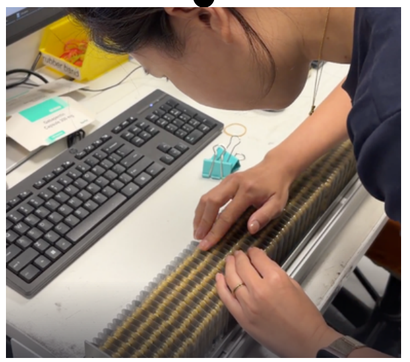
Recount strips and input into DDS-2 System

User Journey Map



Read and verify blister information on DDS-2 System

Recount and verify blister count



Summary of User Journey Map

Task	Step 1	Step 2	Step 3
Counting			
Reading information on blister			
Verify information on blister			

Time for Restocking Each Cartridge

Step	Average Time (min)
1	5.52
2	1.25
3	1.32
Total time (min)	8.09

50-60 Cartridges/day ~ 6.74 - 8.09 hrs

For a 50-min cyclical task

By the final <u>20 mins</u>,

↑20%

Response times

By the final <u>12.5 mins</u>,

↑7%

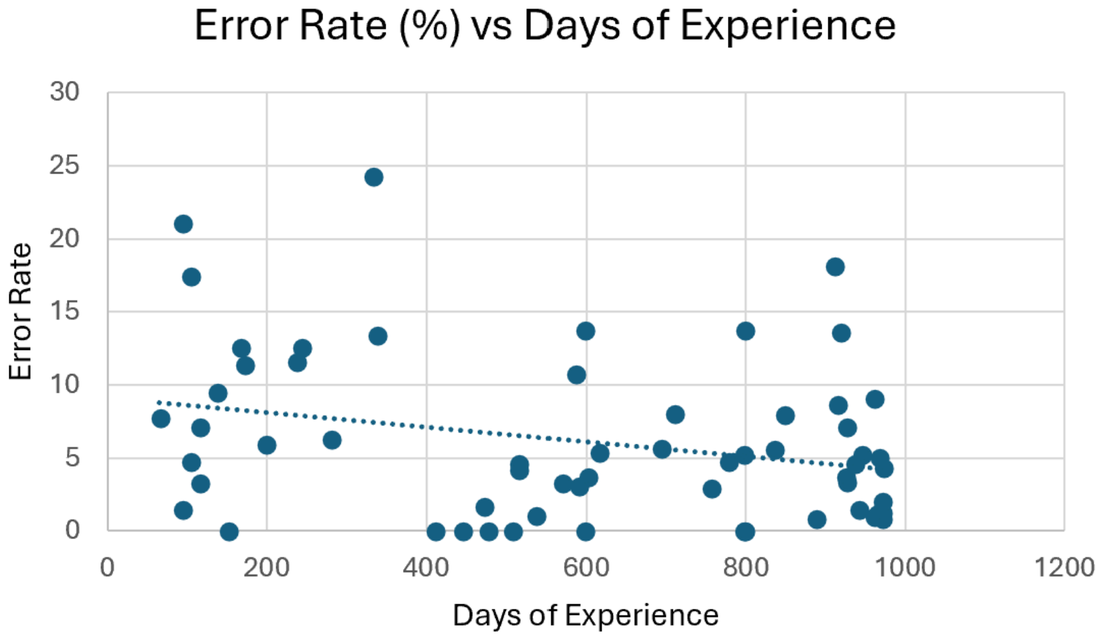
Error rates

Atashfeshan and Razavi, The Journal of Human Factors and Ergonomics Society, 2016

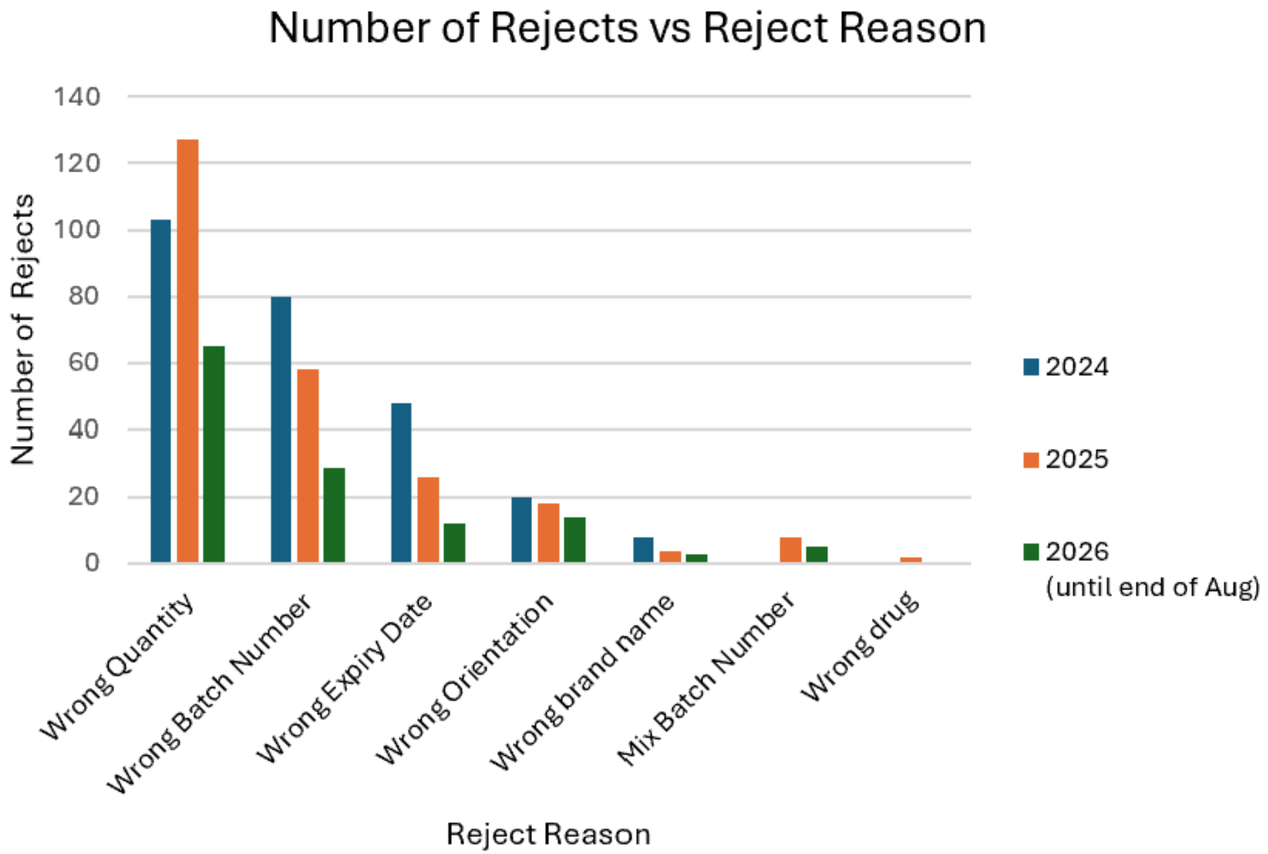
Error Rate by Staff Experience

There was a weak, non-significant negative correlation (ρ = −0.22, p = 0.09) between days of experience and error rate

Experience alone did not explain who made errors. Errors occurred at all experience levels



Staff who loaded < 10 cartridges were excluded (6/66), resulting in a total n = 60.

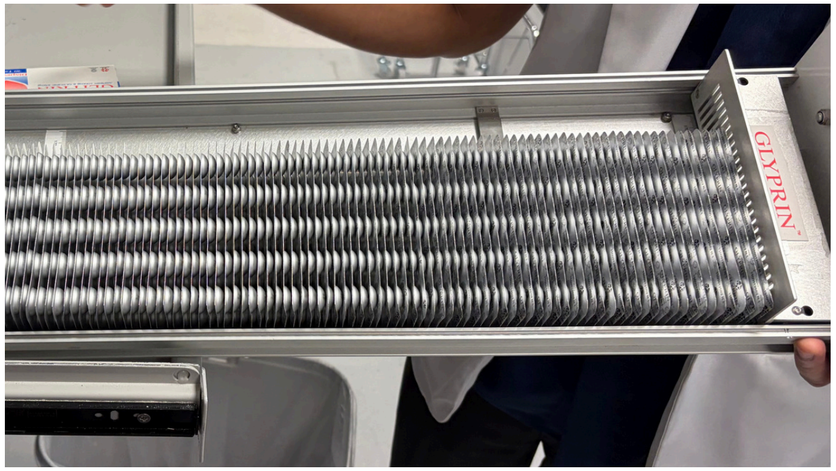


Main Cause of Errors

Wrong Quantity (46.82%), keying in Wrong Batch Number (26.51%), and Wrong Expiry Date (13.65%) are the main causes of errors when loading the DDS

Main Cause of Errors

1



Miscounting

2



Wrong Batch Number or Expiry Date

Potential excess hospitalisation cost when a medication error reaches the patient

≈ S$3,650/ medication error

AND WITH 7 hospitalisation days

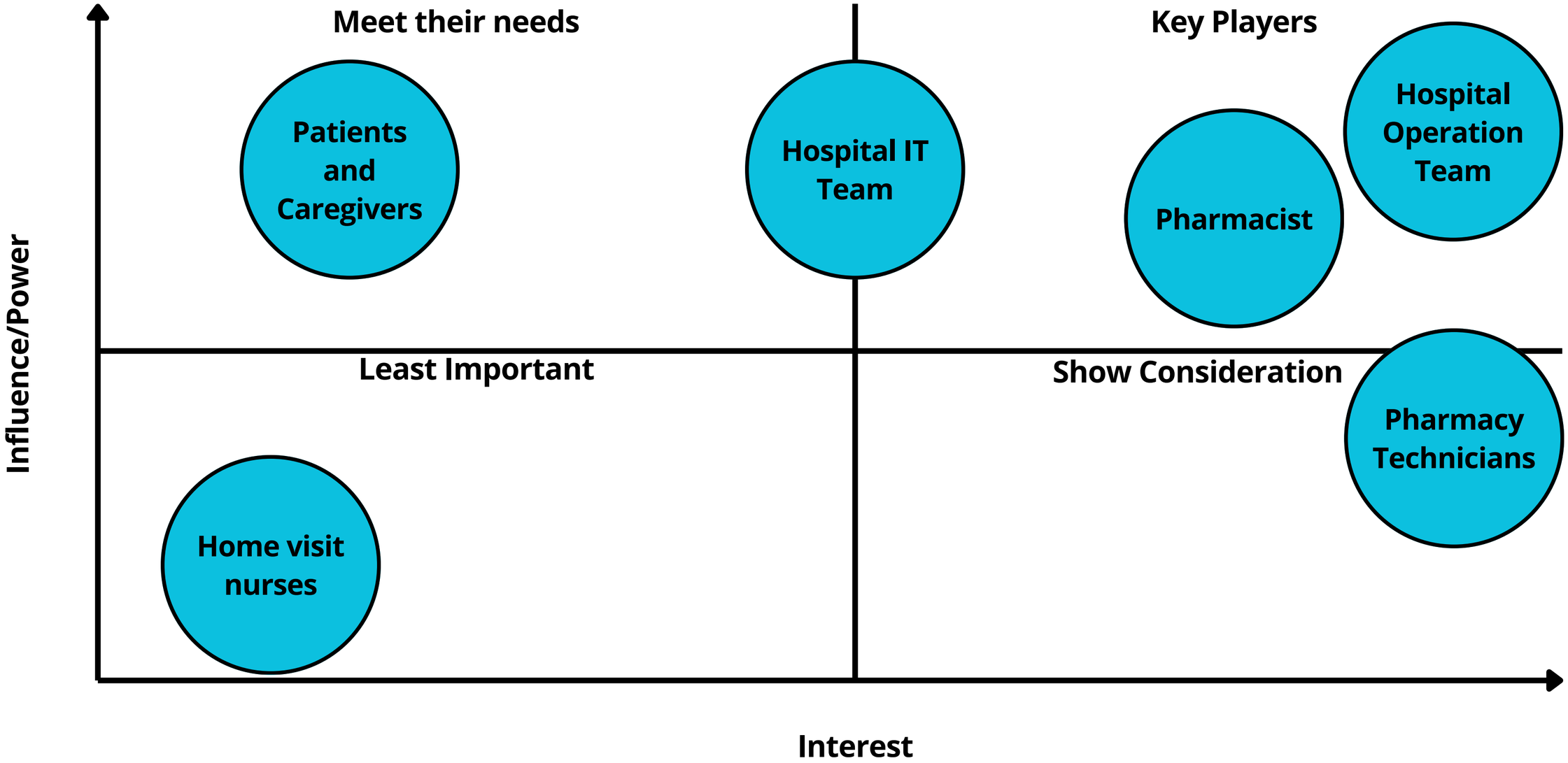
Ng et al., Journal of the American College of Clinical Pharmacy, 2020.

Projected annual manpower savings from pharmacy automation

Up to S$3.3M / year opportunity

NUHS / Centre for Healthcare Innovation, “Inpatient Pharmacy Automation System (IPAS): Redesigning the Medication Supply Process,” 2016.

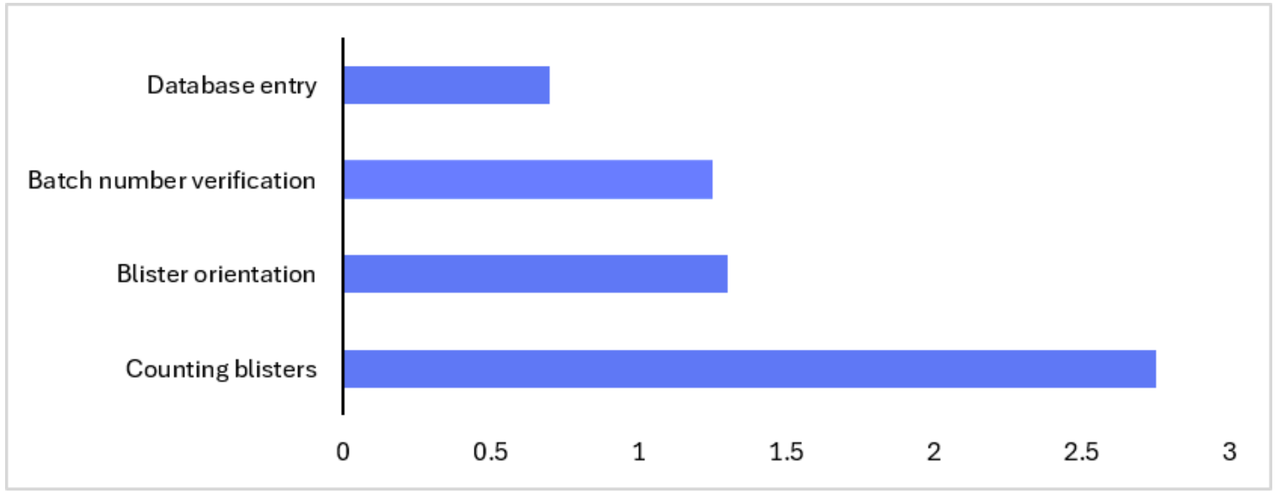
Stakeholder Analysis



User Survey

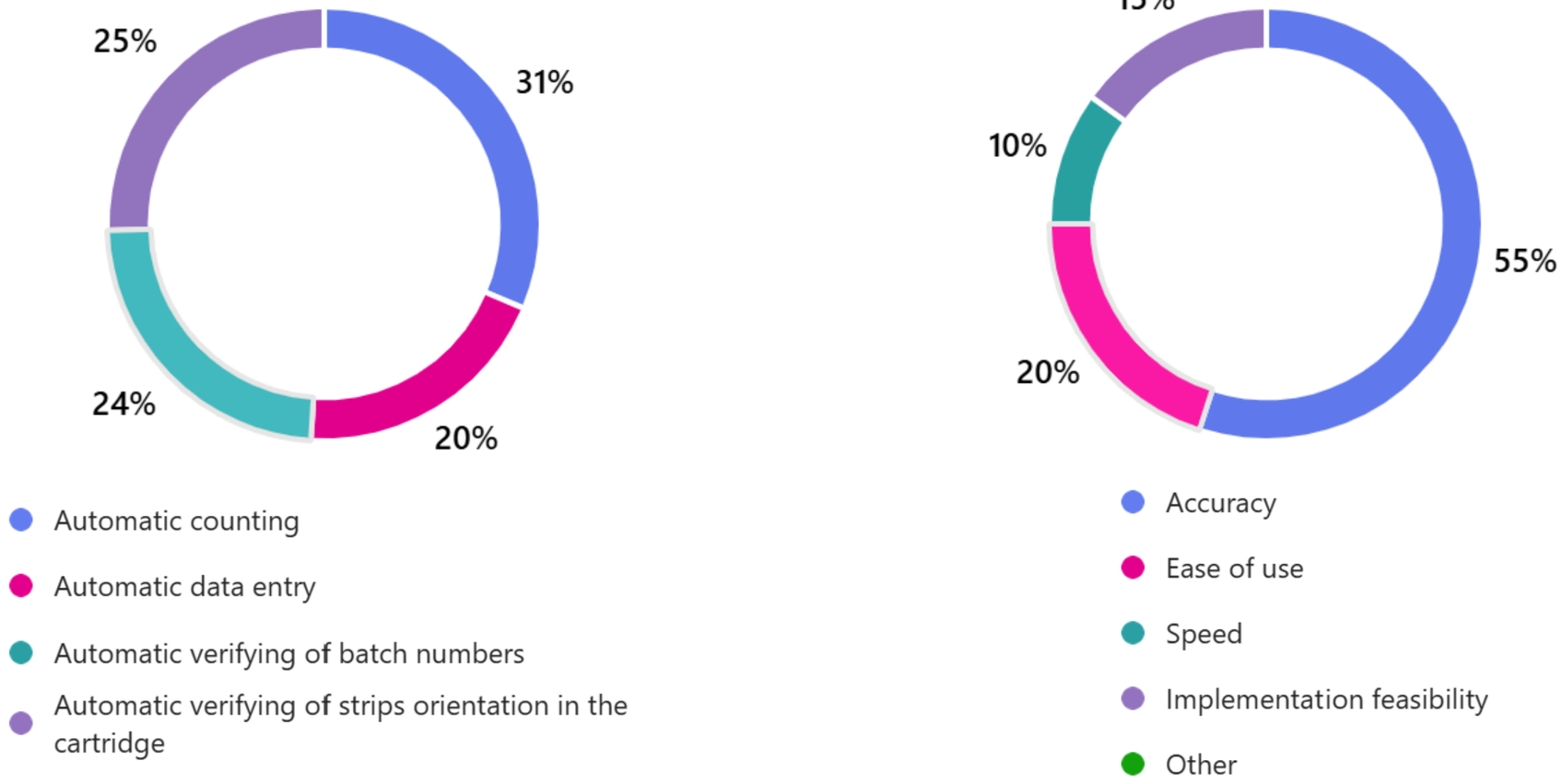
Survey Respondents: 20
Staff involved in DDS-2 medication restocking

Which stages of restocking are the most difficult?



Staff experience a repetitive, high-volume restocking workflow, with counting of strips identified as the most difficult step.

User Survey



Reduce repetitive manual work of counting without compromising accuracy.

Problem Statement

Despite stringent medication-safety requirements, NUH Pharmacy's manual DDS-2 cartridge replenishment workflow is vulnerable to counting and medication verification errors due to human workload, visual limitations, label clarity, and packaging variation.

Design

Must-Have vs Nice-to-Have

Nice-to-Have
Must check orientation of strips in cartridge
Key in drug information and counting on DDS-2 System
Portability/ compact
Verify inputs on DDS-2 System
Easy to understand user interface
Useable by third-parties (i.e. home-visit nurses and patients)

Must-Have vs Nice-to-Have

Must-Have
~99.99% Accurate
Work locally without impacting the pre-existing DDS system
Must automate counting the number of strips
Must not damage medication packaging
Reduce manpower required for the process
Must not take more time than existing replenishment workflow
Must handle different strips packaging and sizes
Feasible with cartridge dimension
Must fit the pharmacy environment (easy to clean, compact, safe around staff)

Design Concept

Create a physical/virtual software tool to assist NUH pharmacists in the loading process by accurately counting and verifying drugs.

Poka-yoke – Developed in the 1960s, it is a Japanese technique that means **“error-proofing”** or **“mistake-proofing”** a process to help operators prevent human errors.

Applying Poka-Yoke to the design should result in a foolproof solution that anyone can easily adopt and that improves cartridge loading efficiency while minimising human error.

Design Specifications

The final design should be able to perform these tasks during the loading process:

1. Count the number of blisters loaded in the machine
2. Verify that all the **blisters' batch numbers, expiry dates, and drug strength** are the same
3. Verify that the **orientation** of all blisters is the same
4. Present the data collected on an LCD screen on the machine

Function	Concept 1	Concept 2	Concept 3	Concept 4	Concept 5
Consume energy	Battery pack	Wall adapter	USB/ Desktop Charging		
Receive user inputs	Button	Touchscreen	Barcode Scanner (scans cartridge?)	Web interface	Automatic cartridge detection
Count blisters	Object Detection Model	Classification on blister edge	Infrared beam	Batch visual inspection	Load Cell
Move / singulte blisters	Spring pusher	Conveyor belt	Passive chute	Manual	Sliding carriage
Detect and extract text	Text detection model (eg. generic YOLO)	Deep Learning & Machine Learning OCR (eg. Paddle OCR/ Easy OCR/	Local LLM (eg. Vision Language Model)	Cloud Based OCR API (eg. Google Cloud Vision API)	DBNet Detector
Process/ Store Data	Raspberry pi	Cloud Strorage	Hospital/ server database	Local database	No long-term storage
Verification Decision	Rule based - Confidence scoring	Multimodal fusion	Human check	Database matching	
Produce visible outputs	LCD Screen	Touchscreen	Web/ app	LED/ buzzer	
Handle Uncertainty	Retry	Reject	Human Escalation	Raise alarm and stop the process	

Handheld Counter **Sorting model** **Sensor with text detection model** **Count+Verify Machine**

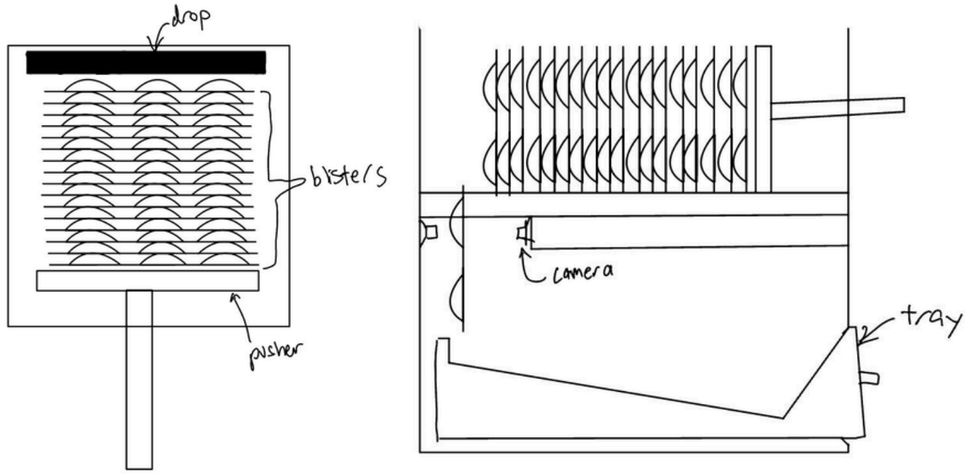
Metrics

Criteria	Concept 1	Concept 2	Concept 3	Concept 4
Accuracy	Depends on the staff with the device	Depends on accuracy of calibration	Depend on accuracy of sensor	Depends on accuracy of calibration and OCR accuracy
Foolproof ability (to human made errors)	Susceptible to human errors	Mostly foolproof	Possible that sensor detects non-blisters	Mostly foolproof
Implementation Feasibility	Depends on training of text detection model	Depends on training of OCR model	Depends on sensor integration and training of text detection model	Depends on training of OCR model
Speed	Slower than the current workflow	Faster than the current workflow	Same Speed/slower than the current workflow	Faster than the current workflow

Prototype 1

AIM: Read the information on the front and back of the blister pack and get the batch info, expiry date, brand name, and potency.

Utilising the Raspberry Pi 5 Camera Module to take pictures of the front and back of the blisters. We train an OCR model on Hugging Face to recognise the text on the blisters.



Prototype 1: Stakeholders Feedback

"Blister thickness and arrangement vary.
How can we prevent multiple strips from falling at once?"

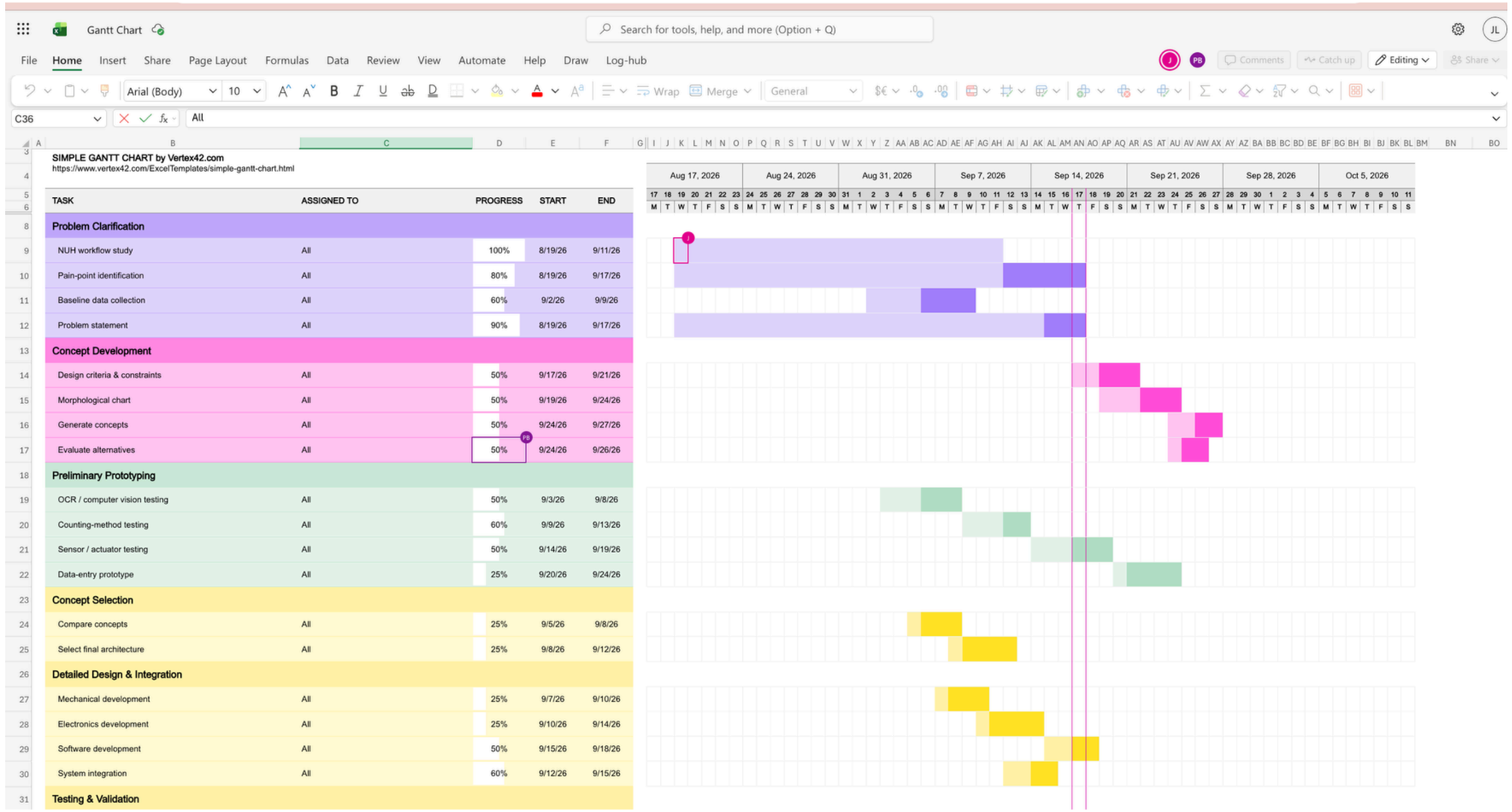
"What if one or two strips are left behind in the tray?
Can a load cell detect leftovers and alert the personnel?"

"Can we replace the collection tray with the actual DDS cartridge for direct loading?"

"We prefer full verification over random sampling, as sampling is too similar to our current workflow."

"Test the pressure applied to the blister strips to ensure the blister does not pop out"

Gantt Chart



TASK	ASSIGNED TO	PROGRESS	START	END
Problem Clarification				
NUH workflow study	All	100%	8/19/26	9/11/26
Pain-point identification	All	80%	8/19/26	9/17/26
Baseline data collection	All	60%	9/2/26	9/9/26
Problem statement	All	90%	8/19/26	9/17/26
Concept Development				
Design criteria & constraints	All	50%	9/17/26	9/21/26
Morphological chart	All	50%	9/19/26	9/24/26
Generate concepts	All	50%	9/24/26	9/27/26
Evaluate alternatives	All	50%	9/24/26	9/26/26
Preliminary Prototyping				
OCR / computer vision testing	All	50%	9/3/26	9/8/26
Counting-method testing	All	60%	9/9/26	9/13/26
Sensor / actuator testing	All	50%	9/14/26	9/19/26
Data-entry prototype	All	25%	9/20/26	9/24/26
Concept Selection				
Compare concepts	All	25%	9/5/26	9/8/26
Select final architecture	All	25%	9/8/26	9/12/26
Detailed Design & Integration				
Mechanical development	All	25%	9/7/26	9/10/26
Electronics development	All	25%	9/10/26	9/14/26
Software development	All	50%	9/15/26	9/18/26
System integration	All	60%	9/12/26	9/15/26
Testing & Validation				

Future Implications

These are the possible advancements in the future:

1. Fully portable where home-visit nurses and patients can use to verify the drugs at home
2. Anyone other the pharmacists can also help to verify and load the cartridges with the design solution as its is 99.99% accurate
3. Self learning computing system that can access the database and stay updated with the lastest changes or recalls of drugs
4. Be used in other pharmacys

Thank You

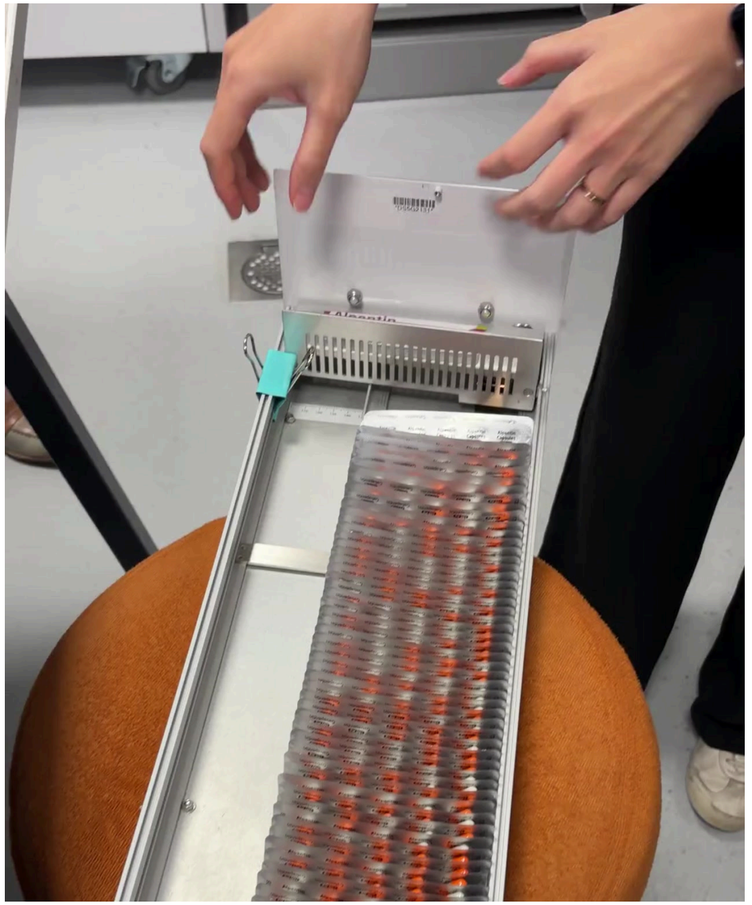
References

1. https://accpjournals-onlinelibrary-wiley-com.libproxy1.nus.edu.sg/doi/epdf/10.1002/jac5.1257
2. https://child.chi.sg/learn/child-collection/projects/inpatient-pharmacy-automation-system-ipas-redesigning-the-medication-supply-process/

Appendix

How cartridge works?





Size and Dimension of Cartridges



There are 4 different sizes of cartridges.

Size and Dimension of Cartridges






Dimension from the Engineer (?) to request from Nicole

Current Workflow

Step 1: Restock



Retrieve required quantities





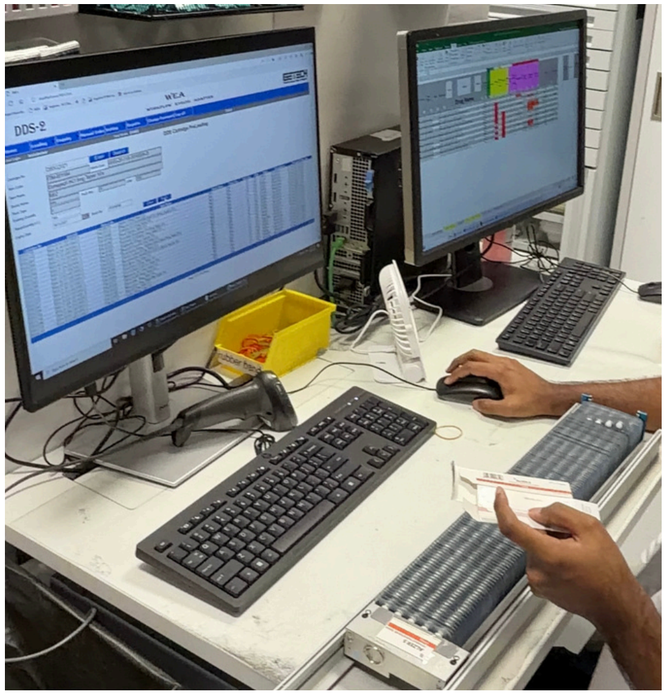
Check two labels within the cartridge

Current Workflow

Step 2: Data Input

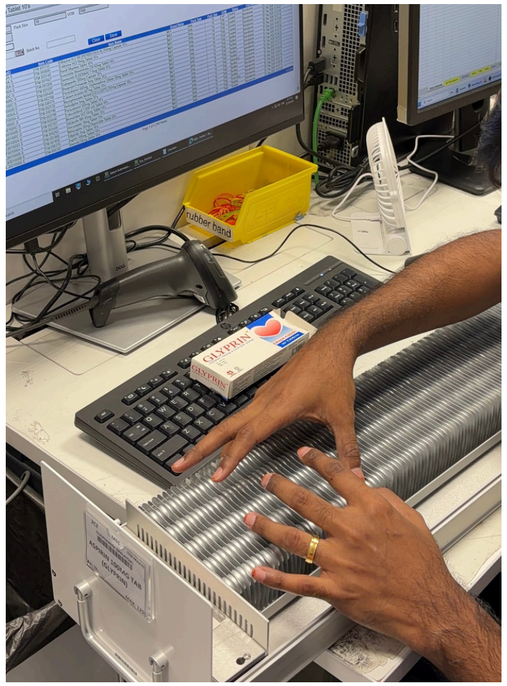


Align strips in right orientation and count concurrently



Key in drug information

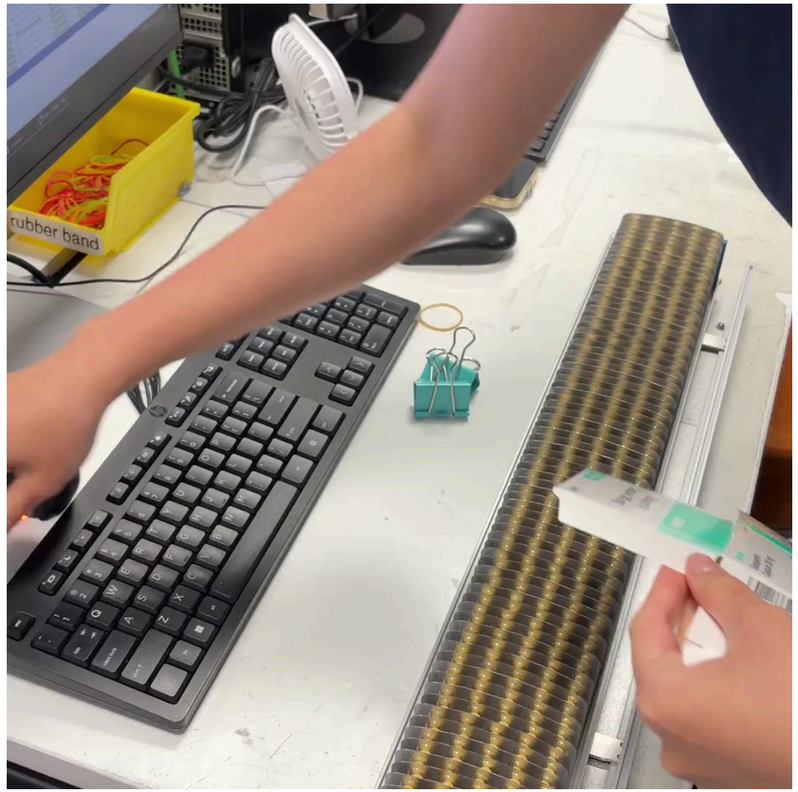
- drug brand
- drug name
- batch number
- expiry date



Recount strips and input into DDS-2 System

Current Workflow

Step 3: Verify

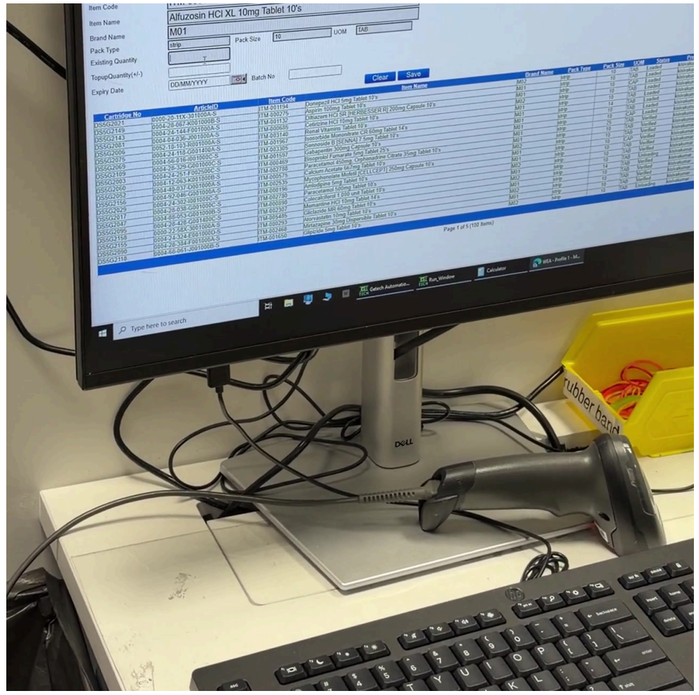


1. **Verify drug information from strip, cartridge and internal vendor list**
2. **Count strips**
3. **Validate whether all previous inputs on DDS-2 System are correct**

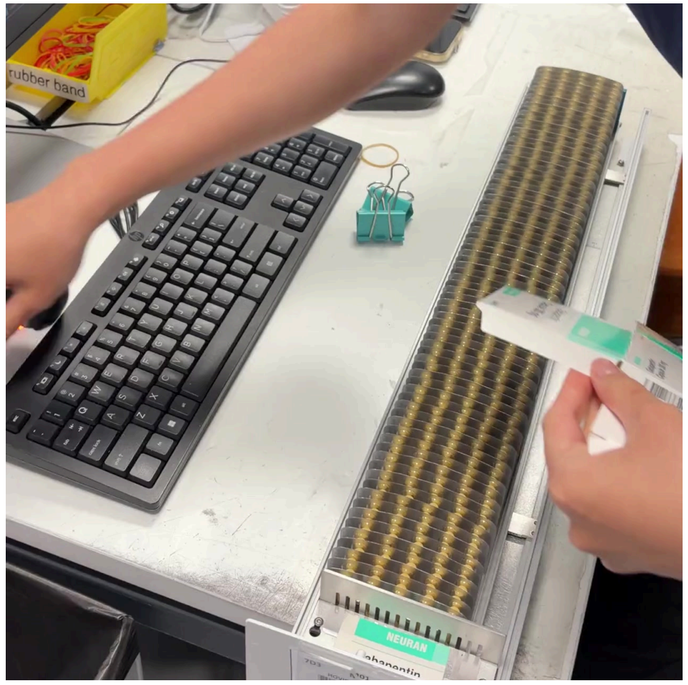
Current Workflow



Restock



Data Input



Verify

Other reasons for errors



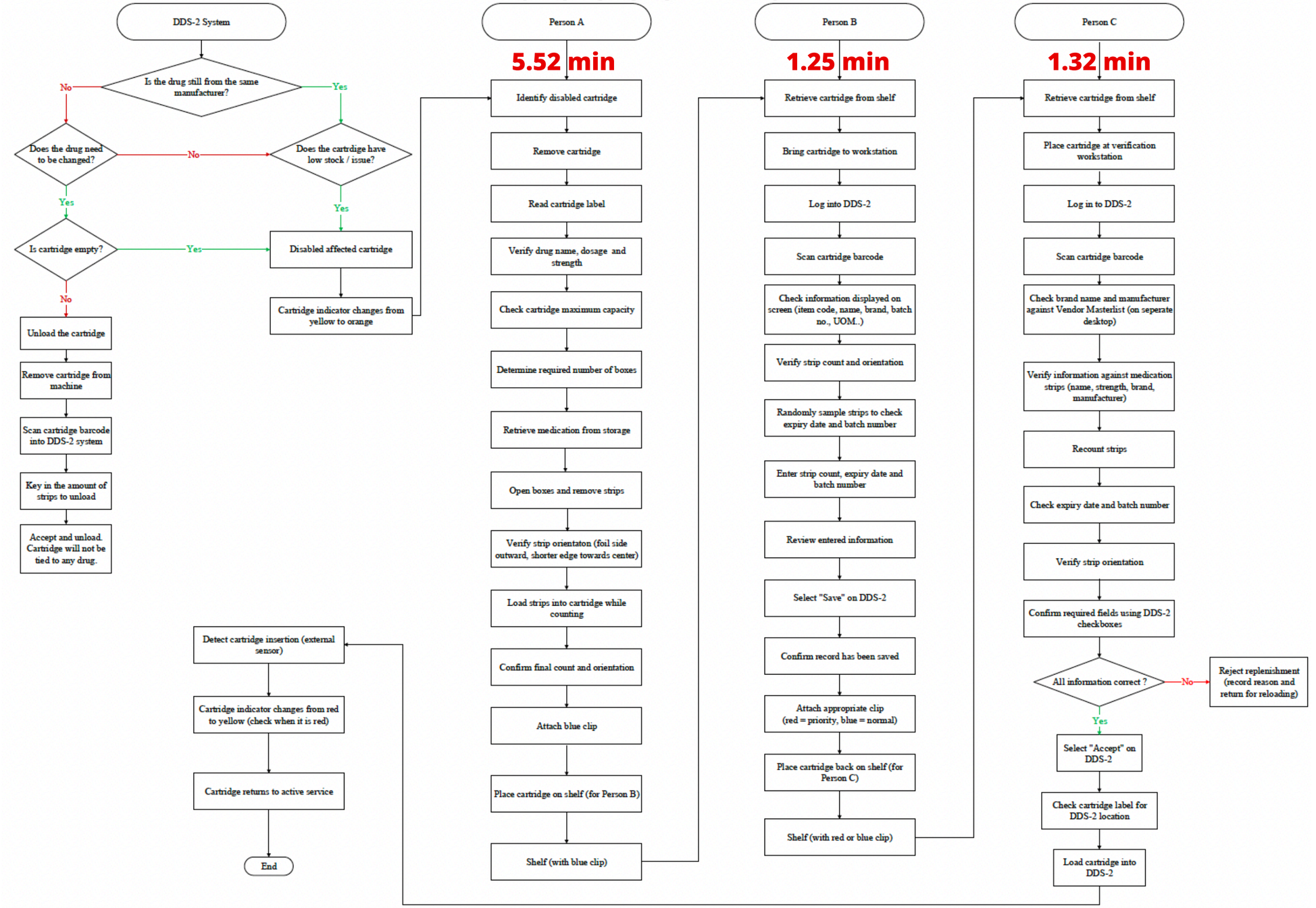
Same drug
different potency





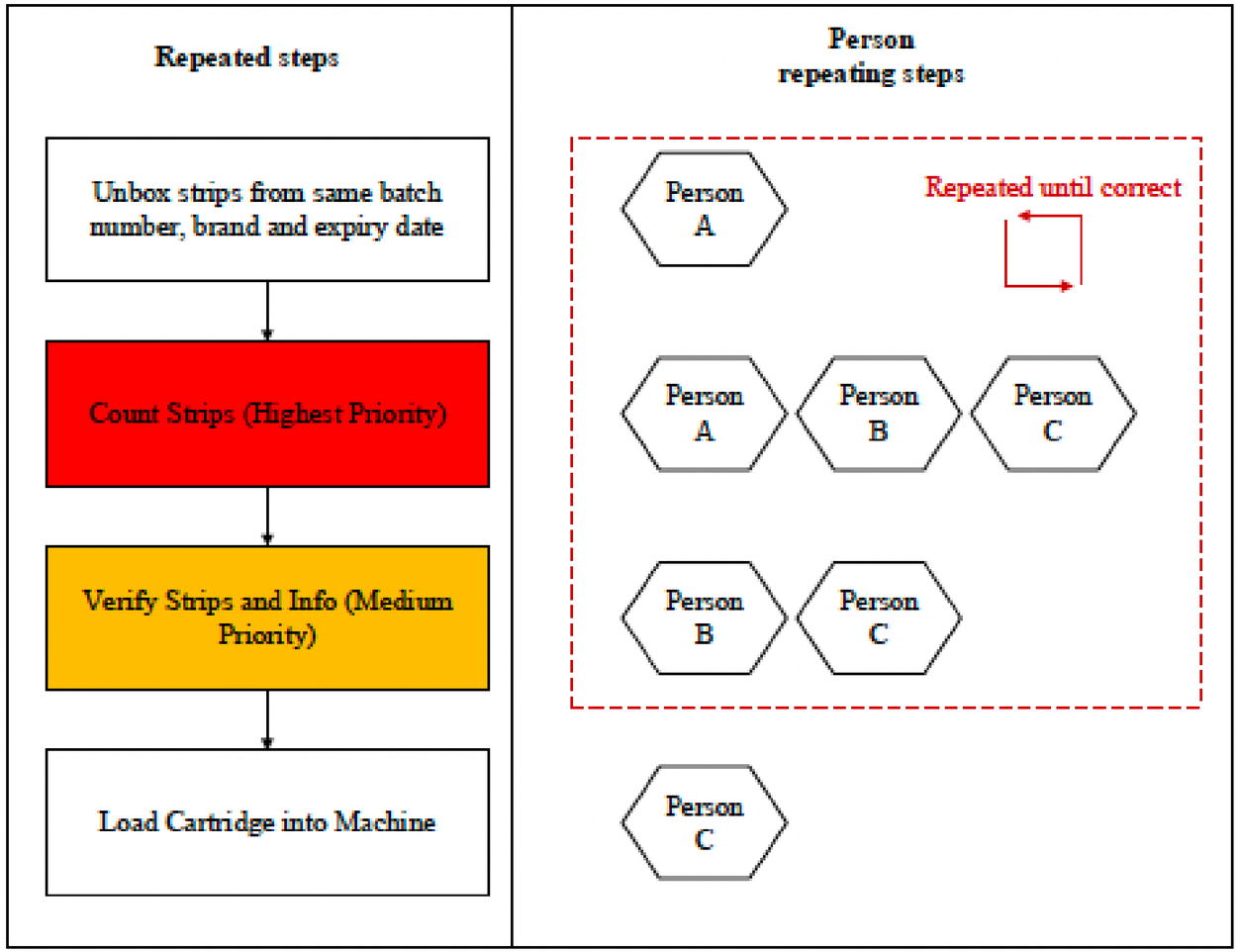
Different drugs that
look similar

Process Flow



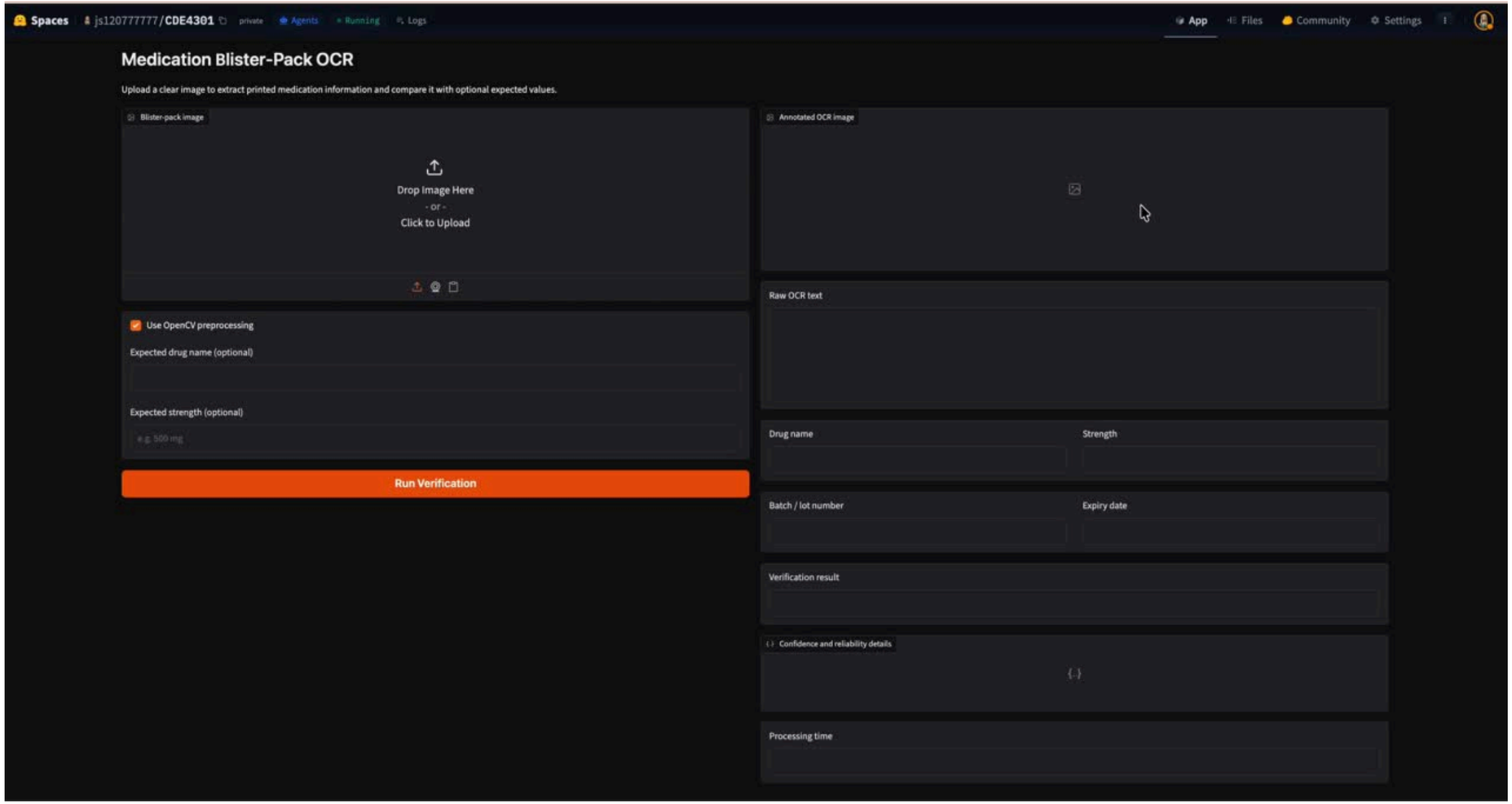
Process Flow

Summary



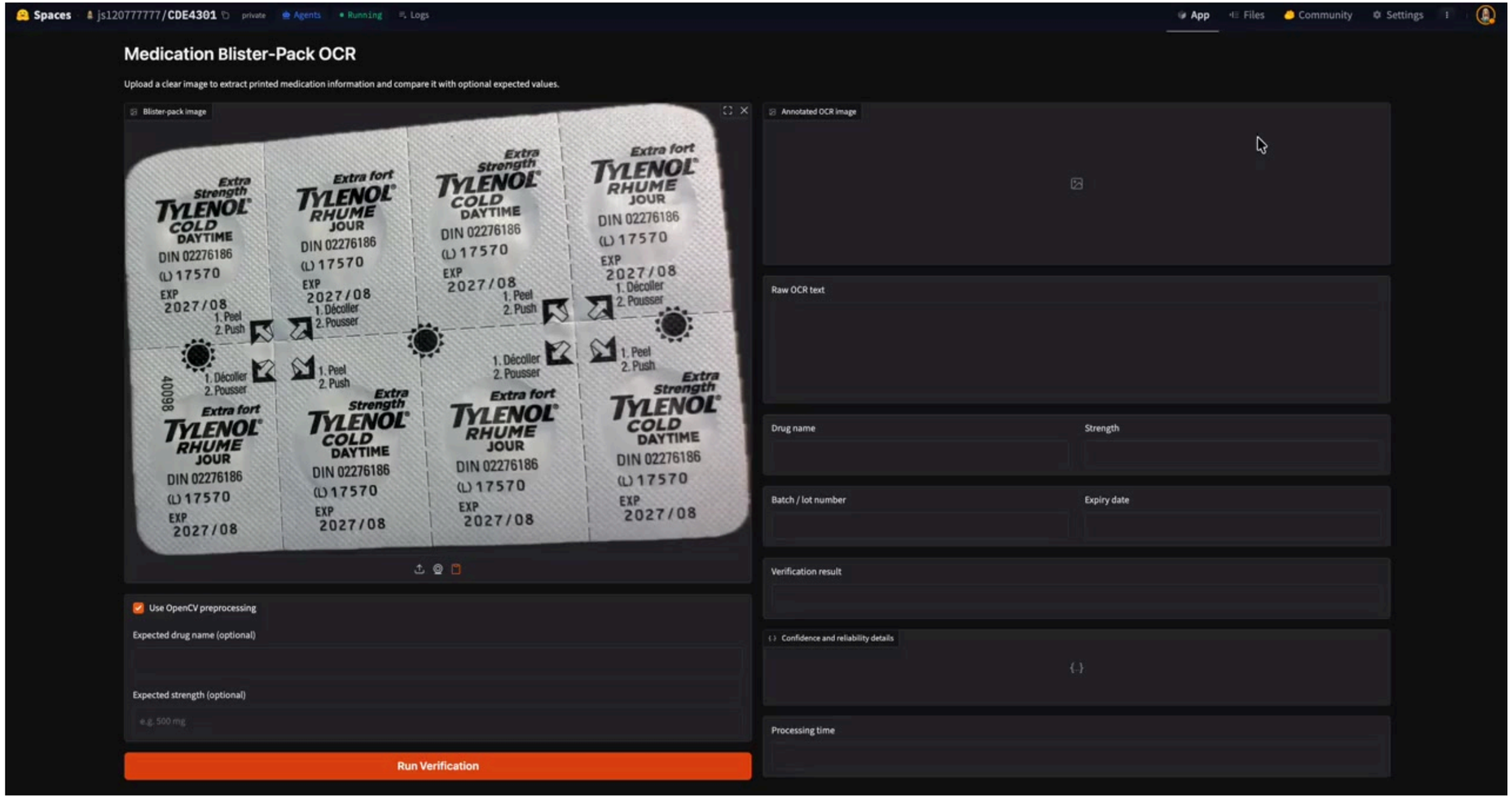
OCR Text Extraction from Blister

1. Text Orientation - Vertically Inverted



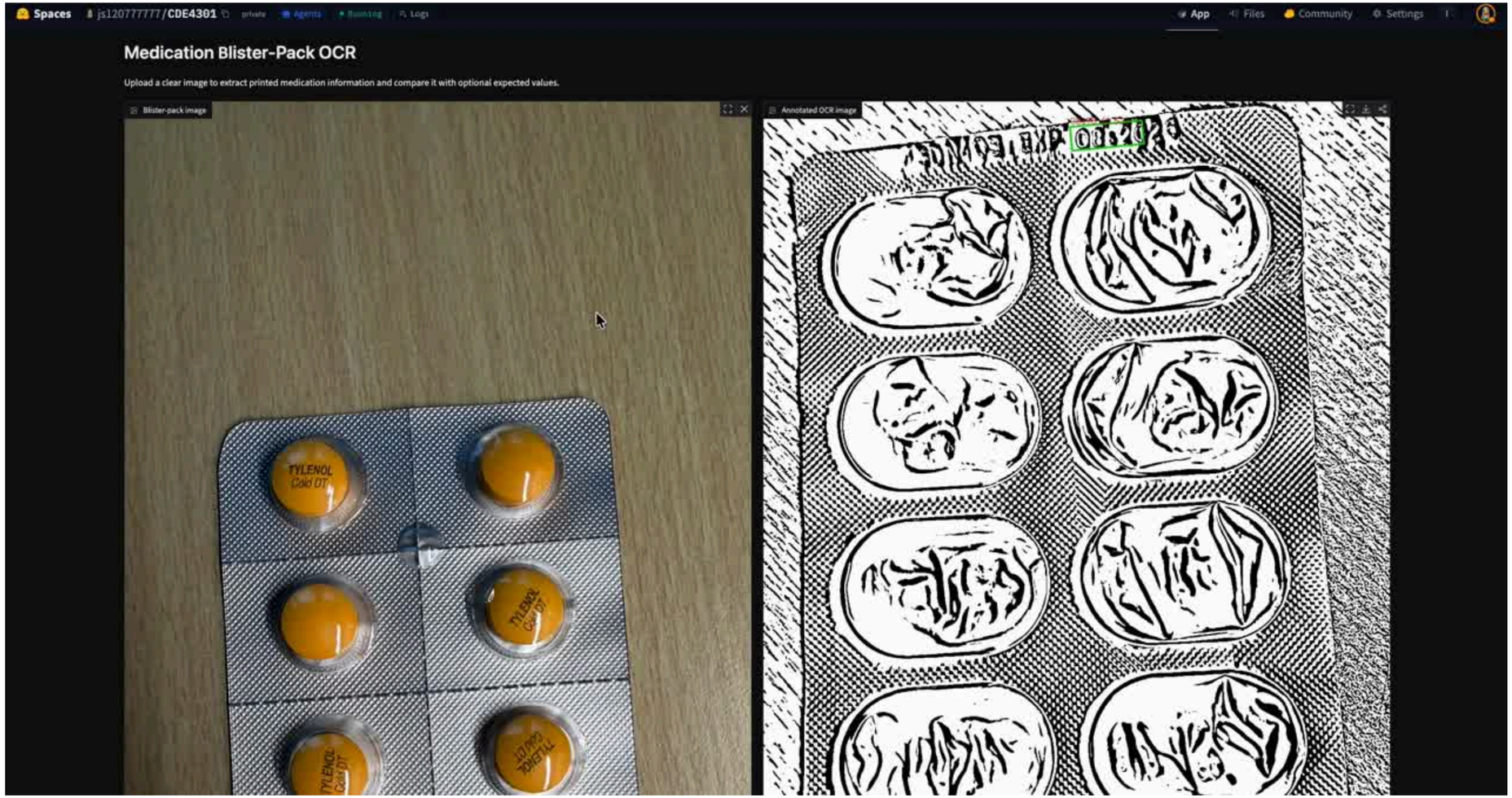
OCR Text Extraction from Blister

1. Text Orientation - Horizontal



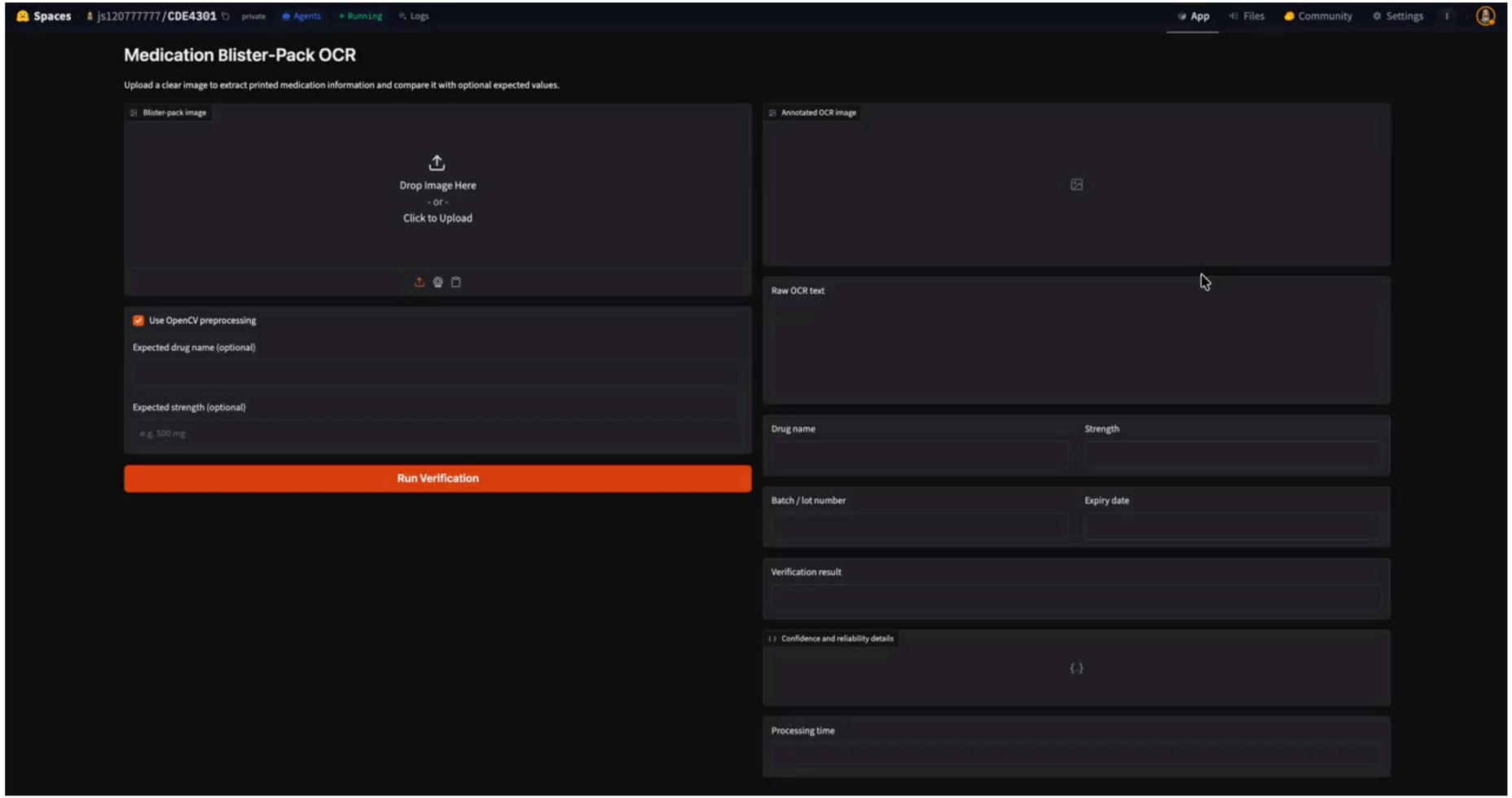
OCR Text Extraction from Blister

2. Text Orientation



OCR Text Extraction from Blister

2. Lightning Effect - Flashed



OCR Text Extraction from Blister

2. Lightning Effect - Without Flash

